SAI TEJA SRIVILLIBHUTTURU

saiteja.srivilli@gmail.com +1 (682) 234-3567 github.com/saitejasrivilli linkedin.com/in/saitejasrivilli

Skills

LLM & GenAI: PyTorch, HuggingFace Transformers, fine-tuning (LoRA/QLoRA, SFT), RLHF/DPO, instruction tuning, vLLM, model serving, evaluation metrics

NLP & Language Understanding: Intent classification, entity extraction, semantic understanding, multilingual NLP, machine translation, prompt engineering, RAG (Pinecone, ChromaDB)

Multi-Agent Systems: LangGraph state machines, multi-agent orchestration, tool use, ReAct reasoning, evaluation frameworks (RAGAS), FastAPI backend deployment

Professional Experience

University of Texas at Arlington — Aug 2024 – Present
LLM Engineer — Texas, US

- Designed and productionized end-to-end **LLM-based** research assistant using **RAG** pipeline indexing 1,000+ academic documents with **Pinecone**, achieving **˜85%** retrieval relevance on 50-query benchmark; presented architecture to 8-person product team and executives.
- Reduced hallucinations **25%** (grounding score 0.73 -¿ 0.91) by implementing **retrieval-aware prompt engineering**, few-shot experimentation, and structured system prompts; coordinated with research team on evaluation methodology.
- Improved long-context reasoning **20%** through **token-aware chunking** and context window management within 4K token limits; benchmarked performance across 3 context lengths (512, 1K, 4K tokens) and documented findings in technical report.
- Deployed scalable FastAPI REST APIs with async streaming and Redis caching, handling **50+ req/s** throughput with **sub-200ms P95** latency; collaborated with DevOps on production monitoring and auto-scaling policies.

DentalScan — Dec 2025 – Feb 2025
Machine Learning Engineer Intern — Remote, US

- Strengthened AI-driven diagnostic workflows by implementing policy-based guardrails, structured output validation, and escalation logic using rule engines across 5 diagnostic tasks; reduced unsafe inference exposure **40%** (0.8% -¿ 0.48% false positive rate); briefed 3 clinical teams on safety improvements.
- Programmed automated evaluation pipelines with metric-based promotion gates (accuracy, precision, F1, recall); prevented **30%** of high-risk model regressions from reaching production across 6 failure categories; collaborated with QA team to define regression detection thresholds.
- Accelerated retraining cycles **25%** (14 days -¿ 10.5 days) via AWS S3- and Lambda-based preprocessing pipelines with version-controlled datasets and automated drift detection; documented pipeline architecture for team handoff.
- Improved deployment reliability by containerizing training and inference workflows with Docker and SageMaker; reduced manual operational effort **35%** and achieved **99.2% uptime** over 2-month period.

Projects

Multi-Agent Research Assistant with NLP Tasks — GitHub

- Designed and deployed **intent classification** system for e-commerce conversations: 5 intent classes (search, purchase, review, complaint, inquiry) with **100%** accuracy; collaborated with product and support teams to define intent taxonomy.
- Programmed **entity extraction** for 7 e-commerce entity types (product, price, brand, color, size, location, quantity); integrated with **multi-agent workflow** for **semantic understanding**.
- Orchestrated **LangGraph** state machine (Researcher -¿ Critic -¿ Synthesizer -¿ Evaluator) with **ChromaDB** vector retrieval and **RAGAS** evaluation (relevancy, faithfulness, coherence), achieving **0.9541 overall** evaluation score.

LLM Fine-Tuning & Alignment: Production System — GitHub — HuggingFace

- **Supervised fine-tuning (SFT)** of Qwen-7B with **LoRA/QLoRA**: **BERTScore improvement 0.780 -¿ 0.855** (+9.6%) on **instruction-following** task with only **0.066%** trainable parameters.
- Multi-GPU distributed training (3x A30) using torchrun and PEFT; deployed model to HuggingFace Hub with production-grade evaluation metrics and trade-off analysis.

Verifiable Alignment: DPO/GRPO Post-Training Pipeline — GitHub

- Implemented full post-training pipeline: **DPO** (direct preference optimization) -¿ **GRPO** (group relative policy optimization with verifiable rewards) -¿ PRM (process reward model) on math reasoning.
- Achieved **12.4% -¿ 30.5% MATH accuracy** on Mistral-7B base model via **alignment** and process-level reward verification using symbolic execution (sympy verifier).
- Implemented DeepSpeed ZeRO-2 for efficient training and **vLLM** for **inference optimization**; tracked experiments via Weights & Biases with detailed metrics.

LLM Brand Safety & Guardrails System — Live Demo

- Crafted production-style multi-modal moderation using **fine-tuned BERT** and CLIP (ViT-B/32) for zero-shot image safety detection, achieving **94% precision** on unsafe content; generating policy-driven verdicts (APPROVE/REVIEW/REJECT).
- Combined text and image risk signals with rule-based escalation logic and explainable probability outputs on Gradio, reducing false positives **28%** vs single-modality baseline.

Education

University of Texas at Arlington – Master of Science in Computer Science — Aug 2023 – May 2025
Andhra University – Bachelor of Technology in Computer Science — Jun 2015 – Apr 2019